

Chapter 4

Maxwell's revenge

4.1 Field equations

We want dynamics for A_μ : a second-order PDE with both Lorentz *and* gauge symmetry. The simplest solution is

$$\partial_\mu F^{\mu\nu} = 0 \quad (4.1)$$

– four equations, one for each value of ν . Take $\nu = 0$:

$$\partial_0 F^{00} + \partial_i F^{i0} = -\partial_i F_{i0} = -\partial_i E^i = 0 \Rightarrow \vec{\nabla} \cdot \vec{E} = 0 \quad (4.2)$$

where we identify the electric field

$$E^i \equiv F_{i0} \quad (4.3)$$

Next take $\nu = j$:

$$\begin{aligned} \partial_0 F^{0j} + \partial_i F^{ij} &= \frac{1}{c} \frac{\partial E^j}{\partial t} + \partial_i F_{ij} \\ &= \frac{1}{c} \frac{\partial E^j}{\partial t} + \epsilon_{ijk} \partial_i B^k = 0 \end{aligned} \quad (4.4)$$

where we identify the magnetic field

$$B^k \equiv \frac{1}{2} \epsilon_{ijk} F_{ij} \quad , \quad i.e. \quad B^3 = F_{12} \quad (4.5)$$

That is

$$\vec{\nabla} \times \vec{B} = +\frac{1}{c} \frac{\partial \vec{E}}{\partial t} \quad (4.6)$$

This looks like Maxwell! (Electromagnetism is the simplest Lorentz invariant coupling of a point particle to a background force field.)

4.2 Sources: charges and currents

Where are the charges and currents? We want, in Gaussian units,

$$\vec{\nabla} \cdot \vec{E} = 4\pi\rho \quad , \quad \vec{\nabla} \times \vec{B} = \frac{4\pi}{c} \vec{J} + \frac{1}{c} \frac{\partial \vec{E}}{\partial t} \quad (4.7)$$

We then need to consider the combined system of point particle with background field PDE. Write

$$\partial_\mu F^{\mu\nu} = -\frac{4\pi}{c} j^\nu \neq 0 \quad (4.8)$$

where j^ν is a (1,0) current – the source – and the factor of $4\pi/c$ is the Gaussian convention. This gives

$$\vec{\nabla} \cdot \vec{E} = \frac{4\pi}{c} j^0 \quad , \quad \vec{\nabla} \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = \frac{4\pi}{c} \vec{j} \quad (4.9)$$

so we identify

$$j^0 \rightarrow \rho c \quad , \quad \vec{j} \rightarrow \text{current density (current per area)} \quad (4.10)$$

4.3 Charge conservation and the Lorentz force

Check: note that

$$\partial_\nu \partial_\mu F^{\mu\nu} = 0 \quad \text{by antisymmetry of } F^{\mu\nu} \quad (4.11)$$

which applied to (4.8) implies

$$\partial_\mu j^\mu = 0 \quad (4.12)$$

That is

$$\partial_0 (\rho c) + \partial_i j^i = 0 \Rightarrow \frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0 \quad (4.13)$$

which is charge conservation. In integral form,

$$\oint \vec{j} \cdot d\vec{a} = -\frac{\partial}{\partial t} \int \rho dV \quad (4.14)$$

Furthermore, the equation of motion of the particle

$$\dot{p}_\mu = \frac{q}{c} u^\nu F_{\mu\nu} \quad (4.15)$$

unpacks into

$$\frac{d\vec{p}}{dt} = q \vec{E} + q \frac{\vec{v}}{c} \times \vec{B} \quad (4.16)$$

– the Lorentz force – together with

$$\frac{d}{dt} \frac{U}{c} = q \frac{\vec{v}}{c} \cdot \vec{E} \quad (4.17)$$

We also now know the transformation of charge and current

$$j^{\mu'} = \Lambda^{\mu'}_\mu j^\mu \quad (4.18)$$

4.4 The other Maxwell equations

Where are the other Maxwell equations

$$\vec{\nabla} \cdot \vec{B} = 0 \quad , \quad \vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t} ? \quad (4.19)$$

They are the Bianchi identity

$$\epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0 \quad (4.20)$$

4.5 Parity and time reversal for $\vec{E}$ and $\vec{B}$

We know $\vec{j} = \rho \vec{v}$ for a point particle, so we expect $j^\mu \propto u^\mu$; hence under parity

$$j^{\mu'} = +P^{\mu'}_\mu j^\mu \quad (4.21)$$

We also know

$$\partial^{\mu'} = P^{\mu'}_\mu \partial^\mu \quad (4.22)$$

which imply

$$F^{\mu'\nu'} = +P^{\mu'}_\mu P^{\nu'}_\nu F^{\mu\nu} \Rightarrow \vec{E}' = -\vec{E} \quad , \quad \vec{B}' = +\vec{B} \quad (4.23)$$

(note the plus sign in $\vec{B}' = +\vec{B}$). Also, under time reversal we expect

$$j^{\mu'} = -T^{\mu'}_{\mu} j^{\mu} \quad , \quad \partial^{\mu'} = +T^{\mu'}_{\mu} \partial^{\mu} \quad (4.24)$$

which imply

$$\vec{E}' = \vec{E} \quad , \quad \vec{B}' = -\vec{B} \quad (4.25)$$

These are like the gravitational and nuclear force fields.

Generally,

$$\partial_{\mu} F^{\mu\nu} = -\frac{4\pi}{c} j^{\nu} \quad \text{and} \quad \dot{p}_{\mu} = \frac{q}{c} u^{\nu} F_{\mu\nu} \quad (4.26)$$

form a coupled, complicated system.